

TEAM ID	SWTID-2026-4030
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	3 MARKS

3.3 – Data Flow Diagram

The Data Flow Diagram (DFD) for the project "Plugging into the Future: An Exploration of Electricity Consumption Patterns" illustrates the flow of data within the electricity consumption analysis system. The system begins with the collection of input data from various sources, including household electricity consumption records, industrial and commercial power usage data, smart meter readings, and region-wise and time-wise consumption details. These inputs provide the foundation for understanding electricity usage patterns across different sectors and locations.

After collecting the data, the system performs several processing activities. These include data collection and integration, data cleaning and preprocessing, consumption pattern analysis, and trend and seasonal analysis. The processed information is then stored in an electricity consumption database containing historical usage records and analytical datasets. This storage ensures that the data is organized and readily available for further analysis and reporting.

The final stage of the system focuses on generating meaningful outputs. These outputs include monthly and yearly electricity consumption reports, region-wise usage analysis, and peak demand identification. In addition, the system provides interactive dashboards and visualizations that help users understand consumption trends more effectively. The insights obtained from the analysis can also be used to forecast future electricity demand and support better energy management and decision-making.

